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CSE 250 Circuits & Electronics

Summer 2026

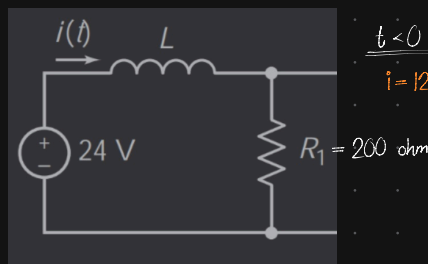
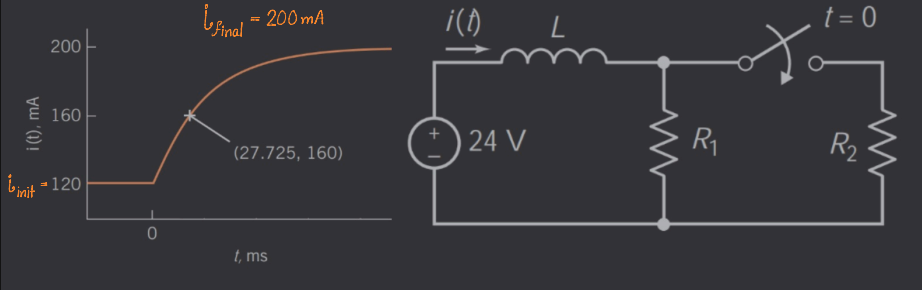
Assignment 03

Section 10A [Tarvin Anjum Aditto]

19th September 2026
10 : 30 p.m.

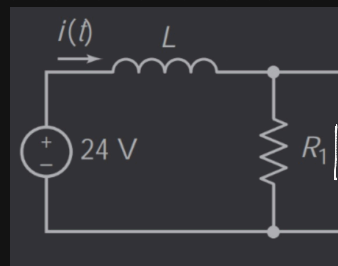
Deadline : 19th September 2026
11 : 59 p.m.

1. The inductor current $i(t)$ has the following plot. Determine the values of L, R_1 and R_2 .



$$\underline{t < 0}$$

$$i = 120 \text{ mA} = \frac{V}{R} = \frac{24}{R} = \frac{24}{200}$$



$$\underline{t > 0}$$

$$i = 200 \text{ mA} = \frac{V}{R} = \frac{24}{R}$$

$$0.2 R = 24$$

$$R = 120$$

$$R_2 = 120 \text{ ohm}$$

$$120 = 200 \parallel R_2 = \frac{200 R_2}{200 + R_2}$$

$$\frac{24000 + 120 R_2}{24000} = \frac{200 R_2}{200 + R_2}$$

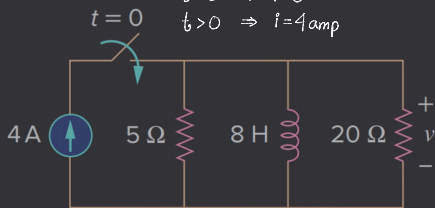
$$24000 + 120 R_2 = 200 R_2$$

$$24000 = 80 R_2$$

$$R_2 = 300 \text{ ohm}$$

2. Find $v(t)$ for $t > 0$

$t < 0 \Rightarrow i = 0$
 $t > 0 \Rightarrow i = 4 \text{ amp}$



$t < 0 \Rightarrow i = 0$
 $t > 0 \Rightarrow i = 4$

$$i(t) = 4 - (4 - 0)e^{-t/2} \rightarrow v(t) = L \cdot \frac{d}{dt}(i(t)) = 8 \cdot (2e^{-t/2})$$

$$R_{eq} = 20 \parallel 5 = 4$$

$$\tau = \frac{L}{R_{eq}} = \frac{8}{4} = 2 \text{ second}$$

$$v(t) = \text{voltage across } 20 \text{ ohm}$$

$$= \text{voltage across our beloved inductor}$$

$$= 16e^{-t/2} \text{ volts}$$